DLR-TomoSAR — Physics-constrained learning

five loss terms · curriculum · the unrolled solver

July 2026

Why constrain the solution space

- The labels are classical fits: the network's ceiling is the label quality — and Act IV showed the network happily leans on statistical shortcuts.
- Physics terms restrict predictions to profiles **consistent with the SAR signal model**:
  - today — as a regularizer: physical quantities recomputed from the predicted profile must match those of the GT profile;
  - next — as self-supervision against the *measured* covariance itself, removing the label ceiling entirely.
- Five terms implemented and theory-tested; every one differentiable end-to-end through the Gaussian-mixture curve reconstruction.

The shared forward operator

All terms use the vertical wavenumber and the steering matrix on the elevation grid:

$$\boxed{k_z^{(i)} = \frac{4\pi\, b_i}{\lambda\, r_0\, s}, \qquad A_{in} = e^{j\, k_z^{(i)}\, \xi_n}}$$

- Geometry resolved from campaign track parameters — not nominal defaults: per-pass perpendicular baselines, look angle, slant range.
- **Per-pixel k_z field** for range/azimuth-varying geometry, built at preprocessing and threaded dataset $\rightarrow$ trainer $\rightarrow$ loss.
- Steering sign settled empirically ($+k_z$: median peak error 0.67 m, within one Rayleigh cell).

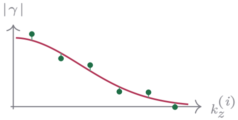
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n)\, e^{jk_z^{(i)}\xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i \left|\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})\right|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences, baseline by baseline.

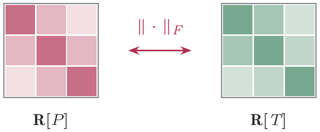
Cloude 2006 (PCT); same Fourier kernel as gamma-Net (Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A}\,\mathrm{diag}(P)\,\mathbf{A}^H \Delta\xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale, entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros & Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

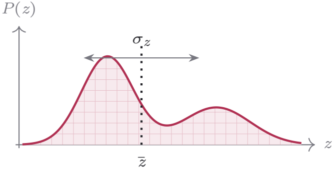
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n)\,\Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \qquad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



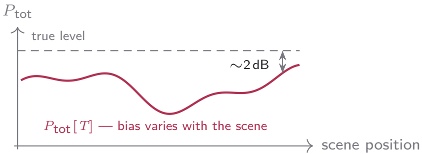
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\,\cdot\,] = \sum_n (\,\cdot\,)(\xi_n)\,\Delta\xi$$

$$\ell_{\text{pow}} = \left\langle \left(P_{\text{tot}}[P] - P_{\text{tot}}[T]\right)^2 \right\rangle$$

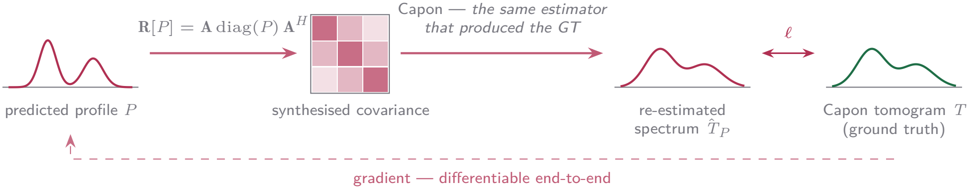


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

$$\hat{T}_P(\xi_n) = \left(\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n)\right)^{-1}$$



- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

Why the physics terms need a curriculum

The physics terms are the most sensitive consumers of the reconstructed profile — and at random initialisation they see only noise:

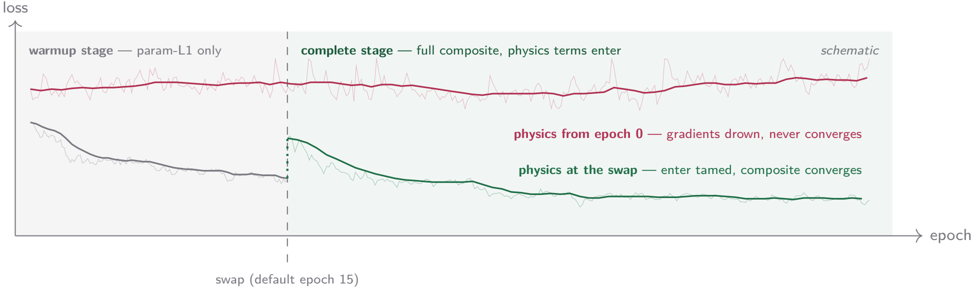
untrained network
predicted profiles are noise

→

physics forward operators
$\mathbf{R}[P]$, Capon solve —
ill-conditioned on unphysical profiles

→

huge, unstable losses
gradients drown the useful signal



Swap mechanics, calibration — and status

Each detail is an anticipated failure mode:

- **Resets at the swap**: LR schedule origin, LR warmup, Adam moments, early-stopping counters — optimizer state calibrated to the old loss is stale under the new one.
- **Checkpoint-baseline reset**: best-loss tracking restarts across the non-comparable loss scales — otherwise the pre-swap best could never be beaten.
- **Stage inheritance**: the complete stage is the base configuration, warmup a derived pre-phase materialised at launch — the stages cannot drift.
- **Loss-scale probe**: every active term run at unit weight over probe batches, IQR-filtered, folded into the per-term weights — physics terms enter at calibrated magnitude, not guessed weights.

Status

Implemented, SAR-theory-verified, geometry wired end-to-end — the single-term $K=2$ sweep is in (next frames); the Capon-cycle arm, the most expensive of the five, is still queued.

The single-term sweep — every family helps low, two collapse high

- At $w \leq 0.05$ **every family beats the no-physics baseline** (0.620) — adding a physics term at gentle weight essentially never hurts.
- **Coherence re-synthesis peaks at** $w{=}0.25$ (0.944, the sweep's best arm) — the theory-ranked #1 term is also the empirical #1.
- **Moments is the robust family**: above baseline at every weight, no collapse anywhere on the grid.
- **Covariance matching and total power collapse past** $w{=}0.25$ — the two absolute-scale terms overpower the parameter loss (curve R^2 drops to 0.72 at $w{=}1$).
- The Capon-cycle arm is not in this sweep — queued, costliest term.

term \ w	0.01	0.05	0.25	0.5	1
coherence resyn.	0.819	0.779	**0.944**	0.609	0.739
moments	0.855	**0.876**	0.792	0.861	0.767
covariance match	0.834	**0.871**	0.578	0.306	0.187
total power	0.789	**0.817**	0.689	0.491	0.144
no-physics baseline			0.620		

runs: dual UNet-skip trunks · set-pred · Hungarian · $K{=}2$ · presence A · hv flips · full stack → both trunks; loss = param-L1 + 0.5 cosine curve + one physics term at weight w; five seeded replicas per arm, held-out test. Cell = grouped score: equal-weight mean of the five metric-group scores (curve error / variance explained / structural similarity / shape / peak location), min-max scaled *within this 21-arm comparison* — rank, not physical units. = below baseline; **bold** = family best; = sweep best. Val losses are not comparable across arms (each optimises a different composite).

The verdict — physics terms buy tail insurance, not headline gains

- **Headline gains are small**: curve R^2 $0.747 \to 0.755$ at the best arm — at the edge of combined seed noise; SSIM moves in the fourth decimal on all three axes.
- **The real effect is the per-pixel tail**: the baseline's mean pixel R^2 is -1.17, wrecked by catastrophic pixels ($p_1 = -14.9$). The descriptor and covariance terms cap the blow-ups and pull the mean positive ($+0.77$ at covariance w=0.05, $p_1 = -0.45$).
- Physics arms also **stabilise seeds**: covariance w=0.05 runs at a fifth of the baseline's curve-metric seed spread.
- Detection and localisation sit **within seed noise** everywhere — the terms shape the profile, not the scatterer set.

	baseline	coh. 0.25	mom. 0.05	cov. 0.05	pow. 0.05
curve R^2	0.747	**0.755**	0.755	0.748	0.751
curve MAE	0.0370	0.0366	0.0365	**0.0365**	0.0366
PSNR [dB]	52.63	**52.78**	52.76	52.65	52.70
SSIM elev	0.9671	0.9674	0.9674	**0.9675**	0.9674
SSIM range	0.9510	0.9516	0.9519	**0.9521**	0.9520
SSIM azim	0.9610	0.9615	0.9615	**0.9616**	0.9615
pixel R^2 mean	-1.17	-0.88	$+0.53$	$\mathbf{+0.77}$	$+0.73$
pixel R^2 p_1	-14.9	-12.4	-2.16	$\mathbf{-0.45}$	-0.87
pixel R^2 min	$-13\,161$	$-12\,209$	$-4\,407$	$-4\,151$	$\mathbf{-4\,065}$
profile cos med	0.9646	**0.9663**	0.9661	0.9658	0.9655
matched recall	0.901	**0.903**	0.902	0.902	0.900
matched precision	0.895	0.892	0.896	**0.899**	0.898
count exact	0.908	0.909	**0.910**	0.909	0.908
μ MAE	1.71	1.72	1.69	**1.65**	1.67
σ MAE	0.774	0.778	0.771	**0.754**	0.761
a MAE	**0.182**	0.182	0.182	0.187	0.184
peak err mean	1.231	1.216	1.221	**1.213**	1.221

runs: dual UNet-skip trunks · set-pred · Hungarian · $K{=}2$ · presence A · hv flips; loss = param-L1 + 0.5 cosine curve + term at w (column = each family's best-scoring weight); five seeded replicas, held-out test, cells seed means. **bold** = best in row; shading keys each block to its like-coloured bullet. Detection / localisation rows all sit inside combined seed noise. matched = Hungarian pairing; μ, σ in elevation units, a in linear amplitude; peak err in elevation units.

Context: where does the physics live?

Three places the SAR signal model can enter a learned inversion — the project now covers all three:

in the labels
model zoo backbones (Act V)
classical fits supervise a generic network — it learns the inversion from scratch; 10^5–10^7 params

→

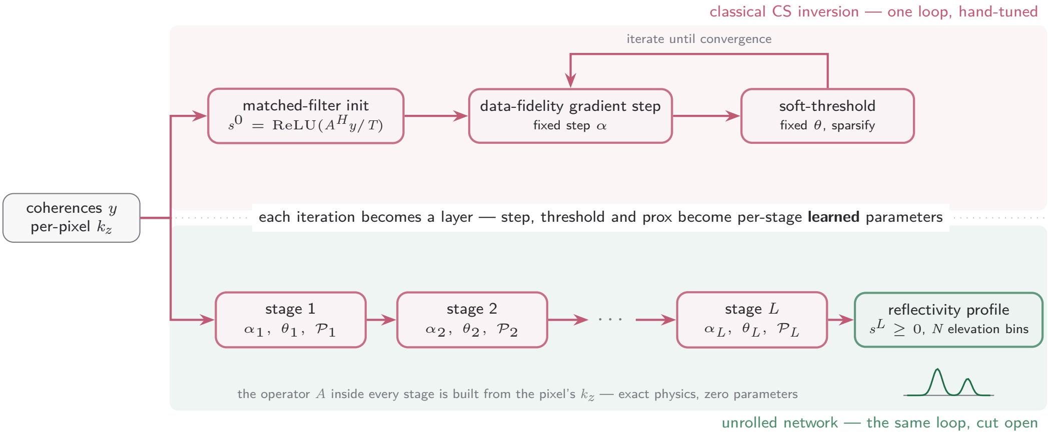
in the loss
physics-constrained terms (Act VI)
predictions must re-synthesise coherences and covariances; the network is unchanged

→

in the architecture
the unrolled solver (next)
the exact operator inside every layer; learned: α_ℓ, θ_ℓ, $\mathcal{P}_\ell$ — ~1.4k params

The unrolled family probes the far end of the axis — **the question it answers**: how much of the zoo's capacity is spent re-learning physics we already know?

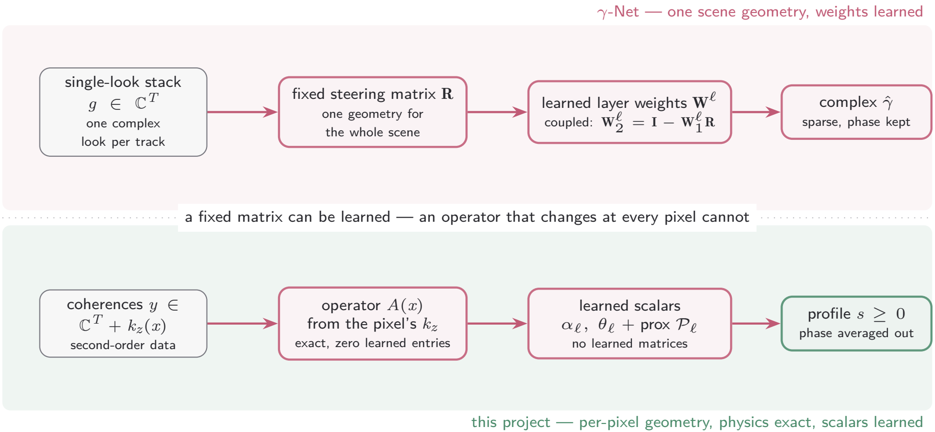
The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \,\mathrm{Re}\big[A^H(y - As^\ell)\big]/L_{\mathrm{lip}}, \qquad s^{\ell+1} = \mathrm{ReLU}\big(\mathcal{P}_\ell(r^\ell) - \theta_\ell\big)$$

γ-Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\mathrm{lip}} = TN\,\mathrm{d}z^2$ normalises the step. Default $L{=}8$: $\sim$1.4k parameters.

Adapting γ-Net — a different unknown, a per-pixel operator



γ-Net inverts $\quad g = \mathbf{R}\,\gamma + \varepsilon, \quad \gamma \in \mathbb{C}^N \qquad$ this project inverts $\quad y = A(x)\,s, \quad s \in \mathbb{R}^N_{\geq 0}$

g: one complex look per track (first-order); y: per-track coherences (second-order) — averaging interferograms cancels the speckle phase, so the recoverable object is the power profile s, not the complex reflectivity γ. ε: measurement noise; T: tracks; N: elevation bins; $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation.

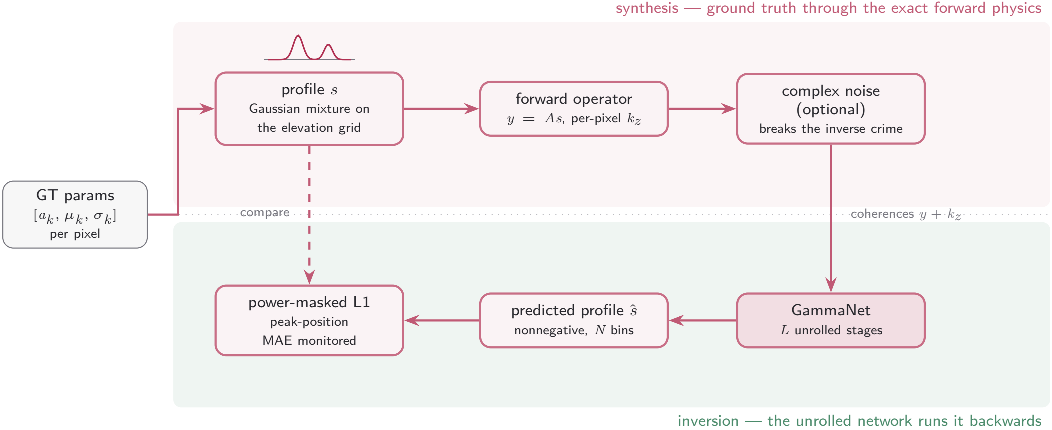
Adapting γ-Net — mechanism by mechanism

mechanism	γ-Net (Qian et al. 2022)		this project	why
iteration	gradient step $\to$ shrinkage, L stages	$=$	kept	the unrolling premise itself
supervision	train on simulated measurements	$=$	kept	the exact operator generates the pairs
unknown	complex reflectivity $\gamma \in \mathbb{C}^N$	$\to$	**power profile** $s \in \mathbb{R}^N_{\geq 0}$	speckle phase averages out of coherences
measurements	single-look stack g	$\to$	**per-track coherences** y	the pipeline is second-order end to end
operator	one fixed $\mathbf{R}$, scene-wide	$\to$	$A(x)$ **from the pixel's** k_z	airborne geometry varies across the scene
learned weights	$\mathbf{W}^\ell$ per stage, coupled to $\mathbf{R}$	$\to$	**none — scalars** $\alpha_\ell,\ \theta_\ell$	a per-pixel matrix cannot be learned
shrinkage	support selection + piecewise-linear function	$\to$	**prox** $\mathcal{P}_\ell$**, then** $\mathrm{ReLU}(\,\cdot - \theta_\ell)$	the prior must respect $s \geq 0$
model order	Bayesian information criterion after inversion	$\to$	**downstream parameter extraction**	the pipeline already fits K scatterers

The contract every change preserves: **the physics stays exact inside the network — only the prior is learned.**

support selection: the largest-magnitude entries bypass the threshold at each stage; piecewise-linear function: γ-Net's learned replacement for soft-thresholding; coupled weights: $\mathbf{W}_2^\ell = \mathbf{I} - \mathbf{W}_1^\ell \mathbf{R}$ halves the free parameters per stage. γ-Net reaches its accuracy at 12 stages; default here $L = 8$. green = exact physics carried by the operator · red = learned · grey = kept from γ-Net.

Training the unrolled model against the exact operator



Phase 1 (implemented): learn to invert the *exact* operator on synthesised coherences. Phase 2 (pending): measured covariances from the stack — no label ceiling.

Isolated family — own registry, trainer and entry point (`train_unrolled`); optimizer splits steps/thresholds from the prox; no AMP, the operator is complex-valued.

Why it exists: the physics-native comparison point — if ∼1.4k parameters of encoded physics match capacity-matched vision backbones, those backbones are mostly re-learning known physics.